JINGTONG HE

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Preferred name: Sophia

EDUCATION

University of Illinois Urbana-Champaign Aug 2023 – Dec 2026 (expected)
B.S. Biochemistry; Minors in Chemistry and Theatre | *GPA: 3.82 / 4.00* *Champaign, IL*

- **Relevant coursework:** Differential Equations; Physical Biochemistry; Brain, Learning, and Memory

Area of Interest: Bioimaging and biomedical image analysis; Neurotechnology; Biomedical Data Analysis; Translational biomedical engineering.

RESEARCH EXPERIENCE

Gritton Lab, University of Illinois Urbana-Champaign Feb 2025 – Present
Undergraduate Researcher (PI: Dr. Howard J. Gritton) *Champaign, IL*

Project 1: The Role of Parvalbumin-Positive Interneurons in Mediating the Expression of Extinction Learning in the Hippocampus (Senior Thesis)

- Studied CA1 PV+ interneuron dynamics during trace eyeblink conditioning (tEBC) and extinction using dual-color calcium imaging in 8 PV-tdTomato-Cre mice; identified transient PV-specific ensemble reorganization at early extinction learning stage.
- Performed stereotaxic viral injection, cranial window implantation, behavioral training, calcium imaging, and histological validation.
- Developed a visual-CS tEBC paradigm in TDT/Synapse with TTL-synchronized stimulus delivery for within-animal sensory-modality comparisons.
- Analyzed population dynamics using PCA and crossnobis distances, with permutation/jackknife inference and drift, cell-sampling, and neuropil controls.
- Simultaneously recorded hippocampal LFP and calcium activity in awake mice and built a Python time-frequency analysis pipeline.
- Quantified conditioned eyeblink responses with DeepLabCut key-point tracking, cross-validated against a custom grayscale-intensity method.

Project 2: Striatal Cholinergic Interneurons Differentially Modulate D1/D2 Medium Spiny Neurons During Movement and Reward (SfN abstract, second author)

- Assisted with animal surgeries supporting striatal calcium-imaging experiments, gaining hands-on experience in experimental neuroscience.
- Helped improve motion-sensor alignment and locomotion tracking in the virtual-reality setup, building on the existing Teensy microcontroller and C++ codebase.
- Contributed to Python-based processing, statistical analysis, and visualization of calcium-imaging and behavioral data to investigate movement- and reward-associated activity in striatal D1-MSN, D2-MSN and cholinergic interneuron populations.

Chan Lab, University of Illinois Urbana-Champaign Oct 2023 – May 2024
Undergraduate Researcher (supervised by Dr. Anuj Yadav) *Champaign, IL*

- Executed multi-step organic syntheses; purified products by flash column chromatography and confirmed identity and purity by TLC and ^{1}H NMR.

CONFERENCE ABSTRACTS AND PRESENTATIONS

- Yuan, B., **He, J**., & Gritton, H.J. (Nov, 2026). Striatal Cholinergic Interneurons Differentially Modulate D1/D2 Medium Spiny Neurons During Movement and Reward. Society for Neuroscience (SfN) Annual Meeting.
- **He, J**., Yuan, B., & Gritton, H. J. (Apr 30, 2026). The Role of Parvalbumin-Positive Interneurons in Mediating the Expression of Extinction Learning in the Hippocampus. Poster presented at the Undergraduate Research Symposium, University of Illinois Urbana-Champaign, Urbana, IL.

HONORS AND AWARDS

School of MCB Summer Undergraduate Research Fellowship (SURF) Mar 2026
Sponsored in part by Dr. Keith Westcott.

Dean's List, University of Illinois Urbana-Champaign Fall 2025

INDUSTRY EXPERIENCE

INNOBIO Corporation Limited Jun 2025 – Jul 2025
Market Research Intern *Dalian, China*

- Assessed the market opportunity for the company's drug microencapsulation technology, which enables sustained release, in pet nutrition, covering existing pet products and competitors, market size, government funding, and demand for sustained-release delivery.
- Reviewed current scientific literature on microencapsulation and sustained-release formulations to evaluate technical feasibility and differentiation against products on the market.
- Designed and fielded user surveys evaluating experiences, preferences, and perceptions of competing products, and analyzed the responses.
- Synthesized findings into written reports and strategic recommendations supporting evidence-based marketing and product decisions.

TEACHING, MENTORING AND OUTREACH

Siebel School of Computing and Data Science, UIUC Jan 2024 – May 2025
Tutor *Champaign, IL*

- Mentored students weekly through online help sessions and one-on-one discussions in Kotlin and computer science fundamentals, adapting explanations to a wide range of skill levels.
- Contributed to an Android course-rating application in Kotlin: remote-server data fetching via JSON parsing and helper classes, dynamic sorting and filtering of course listings, and favorite-marking with SharedPreferences for local persistence.

Oxbridge Summer Camp, CamExpress (JianTeng) Jun – Jul 2023; Jun – Aug 2024
Tutor and Research Assistant *Shanghai, China*

- Designed and taught hands-on organic chemistry laboratory sessions for high school students (caffeine extraction, acetaminophen and aspirin synthesis).
- Guided students through experimental data analysis and interpretation, connecting bench technique to underlying chemical principles; coordinated lesson delivery and lab safety with a team of TAs.

University of International Business and Economics, International Summer School (UIBE ISS) Jan 2025 – Present
College Representative (part-time campus ambassador) *Champaign, IL*

- Led on-campus publicity and recruitment for UIBE ISS at UIUC (6 info sessions; 26 enrollments attributed to my outreach); partnered with student organizations, liaised with UIUC departments on credit transfer, and advised prospective students one-on-one.

TECHNICAL SKILLS

- **Systems Neuroscience:** single-photon calcium imaging; hippocampal LFP recording; virtual-reality behavioral paradigms; locomotion tracking
- **In Vivo & Histology:** stereotaxic viral injection; craniotomy and cranial window implantation; transcardial perfusion; sectioning; immunostaining; fluorescence microscopy.
- **Neural Data Analysis:** calcium imaging analysis; neural population analysis (PCA, crossnobis/RSA); electrophysiology analysis (spectral and time-frequency); behavioral tracking (DeepLabCut).
- **Programming:** Python, R, Kotlin
- **Hardware & Instrumentation:** TDT/Synapse data acquisition; TTL synchronization; Teensy microcontroller programming
- **Molecular & Cellular:** cloning, cell culture, PCR, Western blot, ELISA
- **Organic Chemistry:** multi-step organic synthesis, flash column chromatography, TLC, ^{1}H NMR.
- **Languages:** English (professional), Mandarin Chinese (native)